Zazr and VAVR Tour

Zazr & VAVR · Replace null-config lookup with Option · Build with Waqar

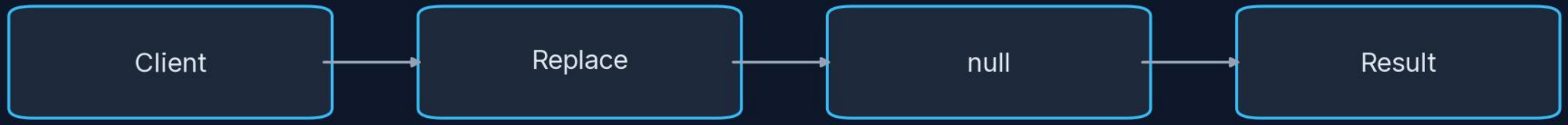
Today's Hook

- Problem: Replace null-config lookup with Option
- Pattern: Zazr & VAVR — earned from the pain, not trivia.
- Runnable main on JDK 25 with captured stdout.
- Naive → solution → failure paths → trade-offs.

The Problem

- Replace null-config lookup with Option
- Naive design shows the cost clearly.
- Pattern isolates the force behind a clean boundary.
- Tests cover happy path and injected failures.

Structure



Zazr & VAVR: Replace null-config lookup with Option

Real-World Scenario

- Running example: Replace null-config lookup with Option
- Same names in naive code, refactor, and Demo output.
- No shared StreamForge framing unless it fits.
- Product language first; pattern vocabulary second.

Naive Code

```
package com.buildwithwaqar.patterns.functional.vavrcompare.problem;

import java.util.*;

public final class NaiveOpt {
  public static String describe(String s){ return s==null? "none": "some:"+s; }
}
```

Why Naive Hurts

- Call sites couple to the wrong shape or leak on partial failure.
- Parameter drilling / duplicated envelopes / missing rollback.
- Hard to unit-test without the real dependency.
- New variants require shotgun edits.

Solution

```java
package com.buildwithwaqar.patterns.functional.vavrcompare.problem;

import java.util.*;
import java.util.function.*;

final class OptionCompare {
  static String zazrDescribe(String raw){
    var o = raw==null? Optional.<String>empty(): Optional.of(raw);
    return o.map(s -> "zazr-some:"+s).orElse("zazr-none");
  }
  static String vavrStyleDescribe(String raw){
    // educational stand-in without requiring VAVR API in problem package
    return raw==null? "vavr-none": "vavr-some:"+raw;
  }
}
```

Runnable main

```
package com.buildwithwaqar.patterns.functional.vavrcompare.problem;
import java.util.*; import java.util.function.*;
public final class ProblemDemo {
  public static void main(String[] args) {
    System.out.println("=== VAVR vs Zazr: same Option describe ===");
    System.out.println("--- naive ---"); System.out.println(NaiveOpt.describe(null));
System.out.println("--- compare ---");
System.out.println(OptionCompare.zazrDescribe("x"));
System.out.println(OptionCompare.vavrStyleDescribe("x"));
System.out.println(OptionCompare.zazrDescribe(null));
System.out.println(OptionCompare.vavrStyleDescribe(null));
  }
}
```

Walkthrough Output

```
=== VAVR vs Zazr: same Option describe ===
--- naive ---
none
--- compare ---
zazr-some:x
vavr-some:x
zazr-none
vavr-none
```

Step-by-Step Trace

- Each stdout line maps to a stage above.
- Re-run the main class locally on JDK 25.
- Failure-path lines prove compensations / rejects / missing files.
- Pause and predict the next line before revealing.

Modern Java 25 / Zazr

```
// Java 25 / Zazr direction
// records + sealed + switch
// inject Env / collaborators at the edge
// virtual threads for blocking fan-out
```

Trade-offs

- Clarity vs ceremony — refuse five types for two lines.
- Teaching demos omit production metrics/idempotency keys.
- Prefer simpler language features when they erase the force.
- Know the twin pattern often confused with Zazr & VAVR.

Related Patterns

- Contrast Zazr & VAVR with its neighbor pattern.
- Name the roles from memory in sixty seconds.
- Point to the season map for what comes next.
- Link the GitHub episode folder in the description.

Recap

- Problem → naive → Zazr & VAVR → Demo stdout.
- Failure paths are part of the lesson.
- Teach when NOT to use it.
- Practice: break a test, fix it, re-run the main.